

# Data Analyst Capstone Project Report

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# OUTLINE

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- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix



## 03 Executive summary

WHAT THE DATA SAYS AT A GLANCE

66.0%

JavaScript current use

39.3%

Python desired next

55.6%

PostgreSQL current use

\$130.8k

Highest scraped salary

27,005

Job records

### The signal

- JavaScript, HTML/CSS and SQL lead current use.
- Python is the strongest “next skill” signal.
- PostgreSQL leads both current and desired database use.
- The job sample is directional, not a labor-market census.

### Decision takeaway

**Build durable fundamentals first — then add Python, PostgreSQL, Rust or Go according to the role you want next.**

This is the through-line across the survey, job sample and salary table.

## 04 Introduction

PURPOSE, AUDIENCE AND VALUE

### Question

**Which technologies and skills are most useful to learn next?**

The analysis joins three perspectives:

- what developers use
- what they want next
- what appears in a job sample

### Audience

**Learners, early-career analysts, hiring teams and program instructors.**

### Value 01

**Translate a large survey into an actionable learning path.**

### Value 02

**Separate broad popularity from forward-looking demand.**

Scope note: Stack Overflow's latest published survey in this analysis is 2025; the two IBM lab datasets are smaller teaching datasets.

## 05 Methodology

THREE SOURCES, ONE DECISION FRAME

01

### Collect

Official 2025 survey pages +  
two lab datasets

02

### Clean

Normalize names, parse  
percentages and correct one  
location typo

03

### Compare

Current use vs desired use;  
jobs vs salary

04

### Communicate

Charts, dashboard views and  
implications

Guardrail

**Use the survey for broad developer signals; use the job and salary labs as directional examples, not like-for-like market estimates.**

# The results combine popularity, momentum and employability signals.

The next four sections move from the two lab exercises into the official survey's technology and demographic views.

01

### Jobs

Where the API sample is concentrated.

02

### Salary

Which scraped languages top the table.

03

### Technology

Current use versus desired next.

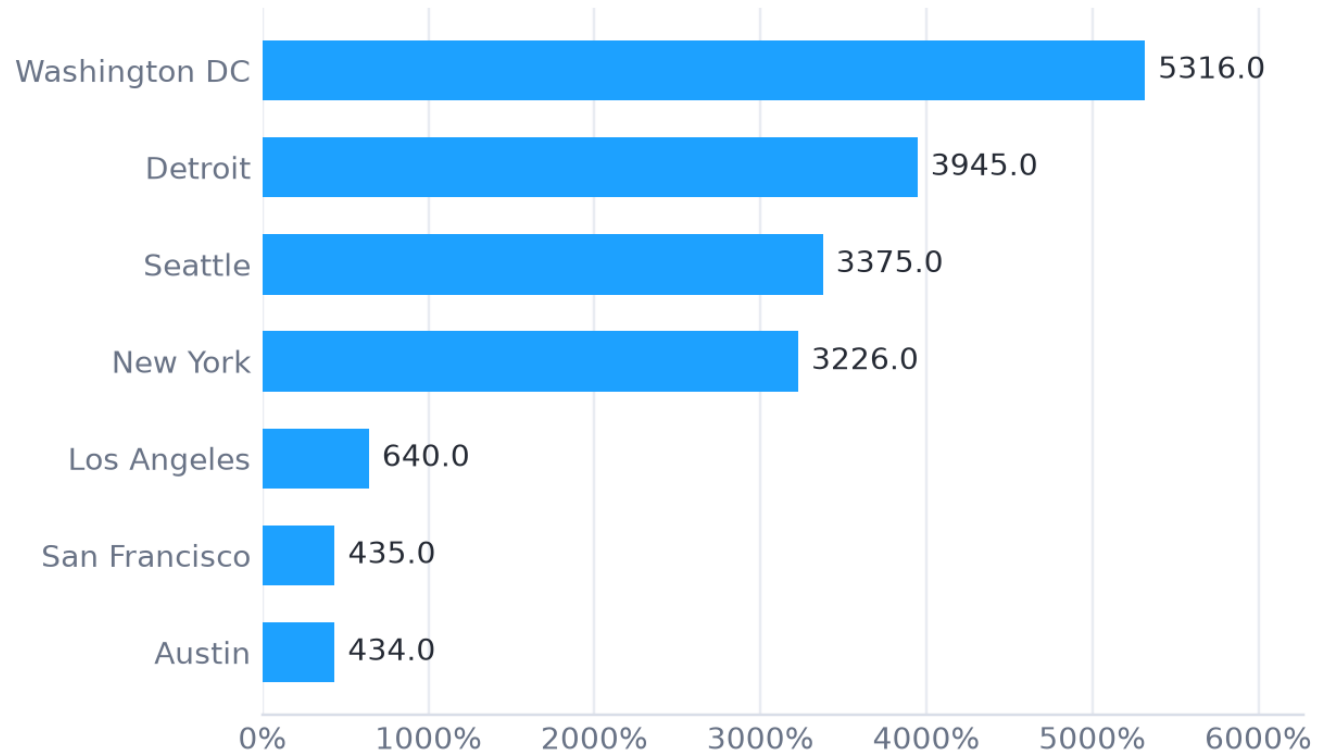
04

### People

Who answered the survey.

## 07 Job postings by location

API LAB RESULT — DESCENDING ORDER



Largest location

**Washington DC • 5,316 postings**

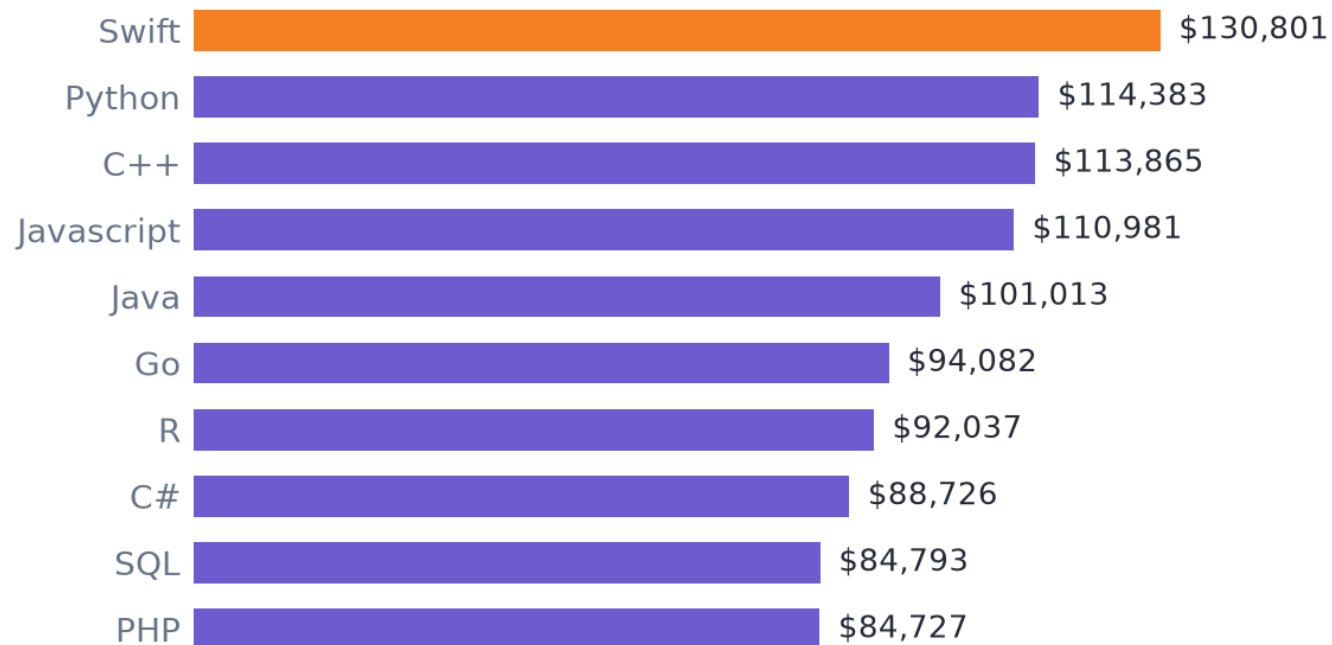
Read this carefully

- Washington DC and Detroit are the largest locations.
- This is a teaching subset, not a live job board.
- The ranking is useful for practicing visualization.

## 08 Popular languages by salary

WEB-SCRAPING LAB RESULT — DESCENDING SALARY

### Average annual salary in the scraped table



Top in the scraped table

**Swift • \$130,801**

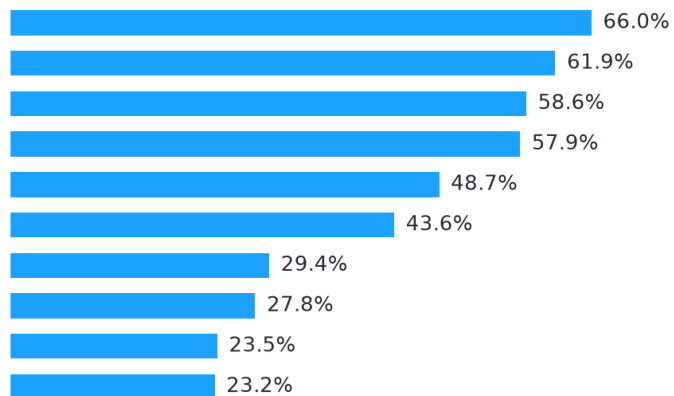
#### Interpretation

- Swift is the highest-paying language in this small table.
- Role, level, geography and industry also shape salary.
- Treat this as a scraping exercise, not a benchmark.

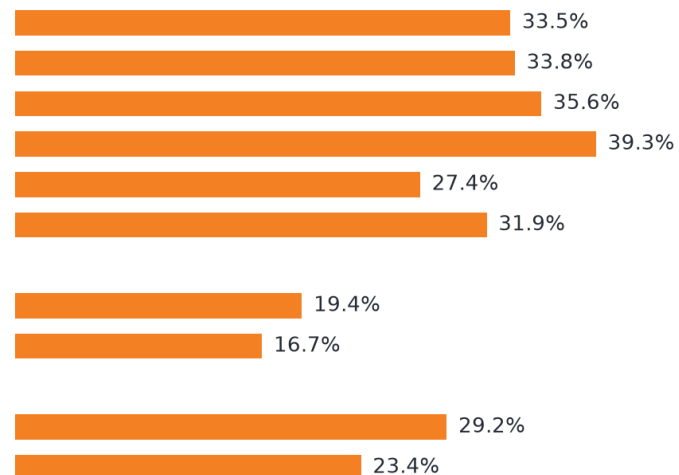
## 09 Programming language trends

CURRENT USE VS DESIRED NEXT-YEAR USE

### Current use



### Desired next



#### Biggest forward signal

**Python leads desired use at 39.3%.**

#### What changes

- Rust and Go rank higher in desired use than current use.
- JavaScript leads current use, not next-skill intent.
- Python connects AI, data and backend work.

## 10 Programming trends — findings & implications

TURNING RANKINGS INTO ACTION

### Findings

#### 01 Core languages still dominate

JavaScript, HTML/CSS and SQL are the top current-use technologies.

#### 02 Python is the clearest growth bet

It ranks fourth in current use but first in desired use.

#### 03 Systems languages are aspirational

Rust and Go have stronger desired positions than current-use positions.

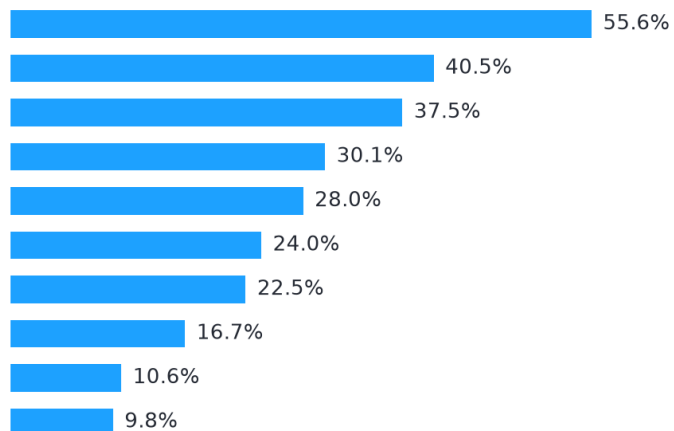
### Implications

- Teach SQL and web fundamentals as the base layer, not as optional extras.
- Use Python projects to connect data analysis, automation and AI.
- Offer Rust or Go as a deliberate specialization after learners have shipped with one mainstream stack.

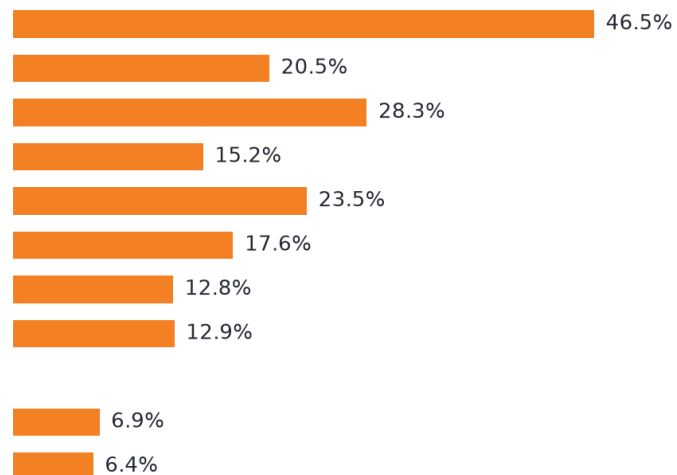
## 11 Database trends

CURRENT USE VS DESIRED NEXT-YEAR USE

### Current use



### Desired next



#### Anchor technology

**PostgreSQL leads both sides.**

#### What changes

- Stack Overflow highlights Redis usage growth.
- SQLite and Redis rise in desired use relative to MySQL.
- PostgreSQL is the safest common denominator for a modern data stack.

## Findings

### 01 Relational skills remain central

PostgreSQL, MySQL and SQLite form the current-use core.

### 02 PostgreSQL has unusual strength

It is both the most used and most desired database.

### 03 Caching and lightweight stores matter

Redis and SQLite show strong forward interest.

## Implications

- Make relational modeling, joins and indexing the baseline.
- Add PostgreSQL as the default portfolio database for learners.
- Teach Redis and SQLite as focused tools with clear use cases, not as replacement fundamentals.

## 13 Dashboard

THREE VIEWS FOR THREE DECISIONS

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### A dashboard should answer a question, not just display a chart.

These tabs separate what is widely used, what developers want next and who is represented in the survey.

TAB 01

#### Current technology usage

What is already common enough to be a baseline?

TAB 02

#### Future technology trends

Where is learning interest moving next?

TAB 03

#### Demographics

Who answered, and what context matters?

## Dashboard tab 1: Current technology usage

WHAT FORMS THE BASELINE

66.0%

JavaScript

55.6%

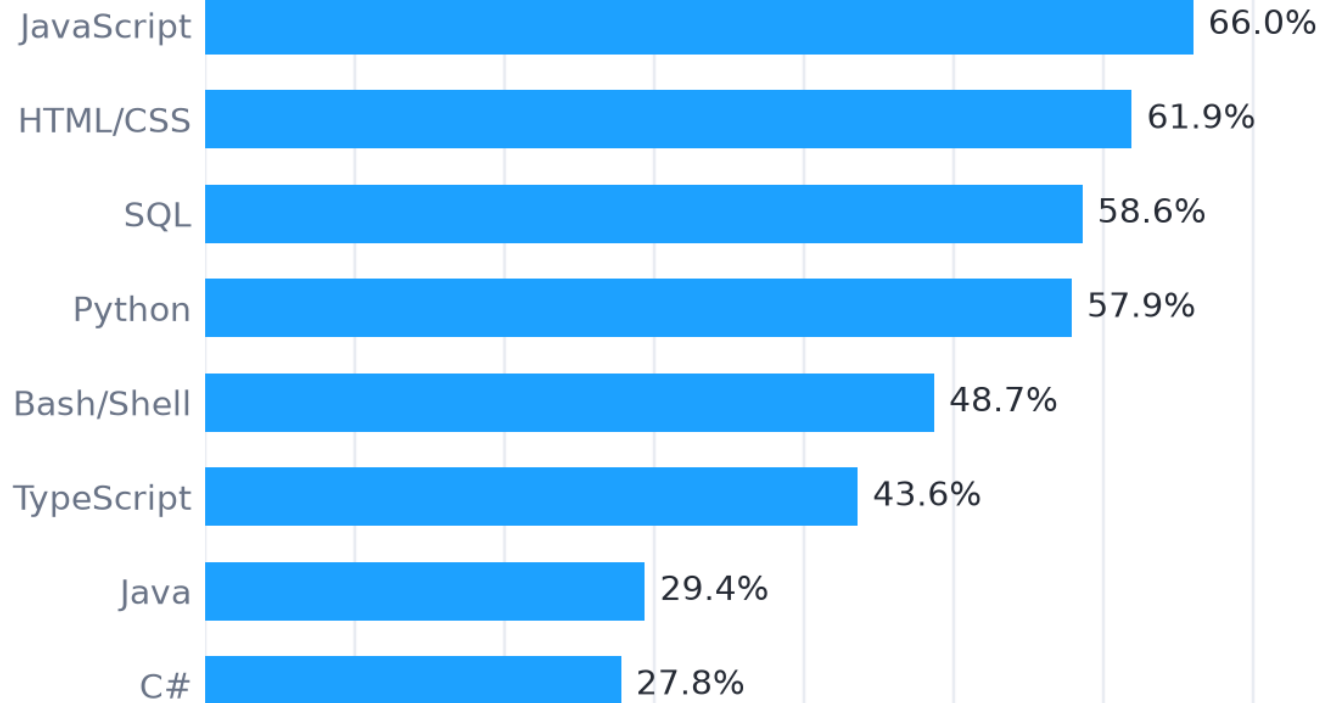
PostgreSQL

58.6%

SQL

57.9%

Python



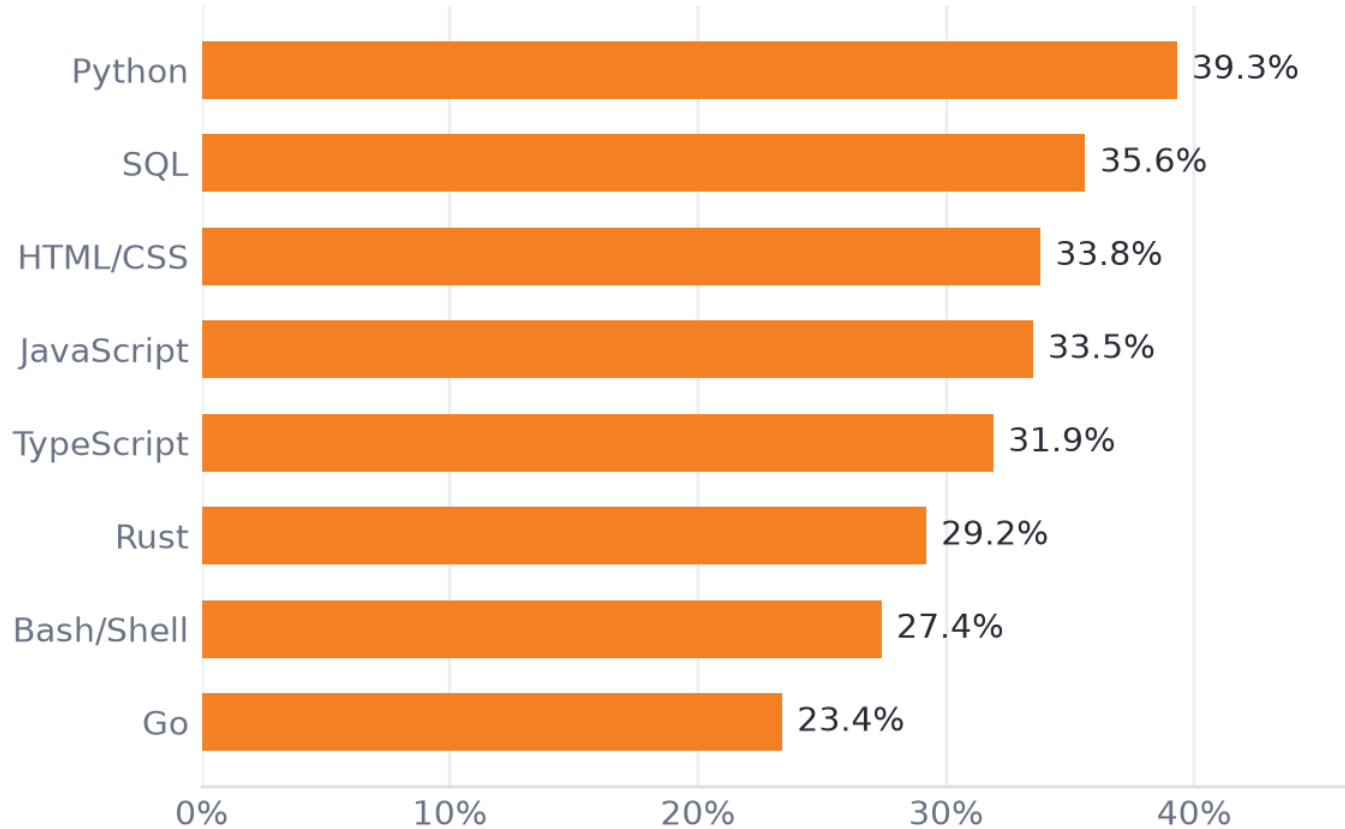
Dashboard insight

**Baseline = web + data:**  
JavaScript, HTML/CSS and  
SQL, with Python close  
behind.

Design implication: teach across the stack.

## Dashboard tab 2: Future technology trends

WHAT DEVELOPERS WANT NEXT



### Strongest next-skill bets

**Python • SQL • Rust**

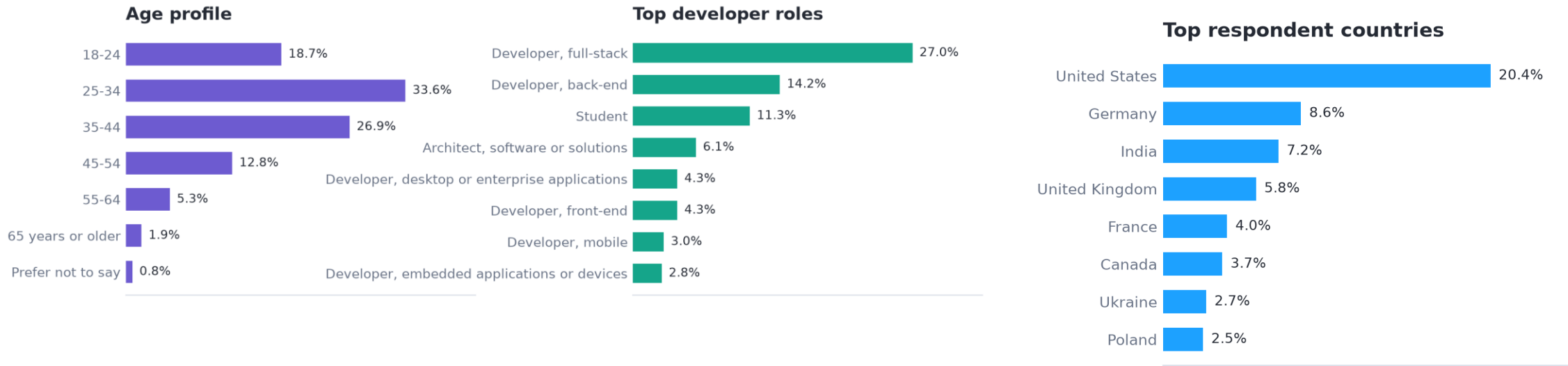
### Dashboard insight

**Interest is not simply following current popularity. Developers are reaching toward Python for AI/data work and toward Rust/Go for systems and backend depth.**

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# Dashboard tab 3: Demographics

WHO ANSWERED THE SURVEY



Context

The largest age band is 25–34 (33.6%). The United States is the largest country group (20.4%).

## 17 Discussion

HOW THE SIGNALS FIT TOGETHER

| SIGNAL         | WHAT WE SEE                                                         | WHAT TO DO                                       |
|----------------|---------------------------------------------------------------------|--------------------------------------------------|
| Baseline       | JavaScript, HTML/CSS, SQL and PostgreSQL are widely used.           | Build fundamentals around web + data.            |
| Momentum       | Python leads desired language use; Redis is highlighted for growth. | Use project-based Python and data work.          |
| Specialization | Rust and Go sit higher in desired use than current use.             | Offer focused pathways after the core.           |
| Context        | The job and salary labs are smaller and older teaching datasets.    | Use them to practice analysis, not to overclaim. |

**Best interpretation: learn what is common, then differentiate with skills that are gaining intent.**

01

## Fundamentals compound

The most-used technologies are still foundational. SQL, web skills and relational databases remain high-leverage.

IMPLICATION

02

## Intent points to Python

Python is the clearest bridge between today's stack and emerging AI, data and automation work.

IMPLICATION

03

## Depth creates differentiation

Rust, Go, Redis and PostgreSQL show where developers are adding depth beyond the baseline.

IMPLICATION

**Recommendation: teach the common core, make Python the applied bridge, and let specialization follow the learner's target role.**

**Start broad. Build depth. Make the next step visible.**

**1**

### **Build the core**

SQL + web fundamentals + one mainstream language

**2**

### **Apply with Python**

Data, automation, APIs and AI-enabled workflows

**3**

### **Specialize deliberately**

PostgreSQL depth, Redis, Rust or Go by target role

**The data does not point to one perfect technology. It points to a sequence: fundamentals → applied Python → role-specific depth.**

## Sources used

- Stack Overflow Developer Survey 2025 — Technology page
- Stack Overflow Developer Survey 2025 — Developers page
- IBM Skills Network Jobs API lab — jobs.json (27,005 records)
- IBM Skills Network Web Scraping lab — 10-language table

## Definitions

- Current use = extensive development work in the past year.
- Desired use = wants to work with the technology in the next year.
- Survey percentages are response shares, not market share.

## Limitations

- The survey is self-reported and global; response coverage varies.
- The job and salary data are teaching datasets, not live market estimates.
- Use rankings to guide questions and learning priorities, not make career choices alone.